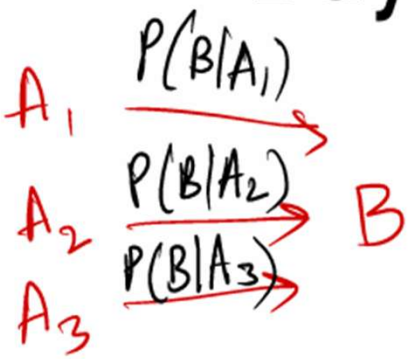
Week 4

Annotations

Bayes' Rule: Motivation

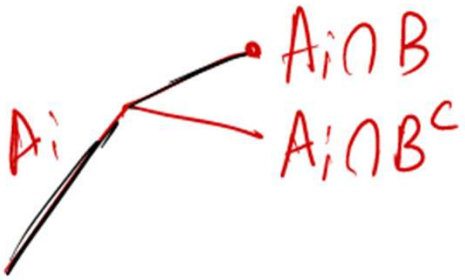


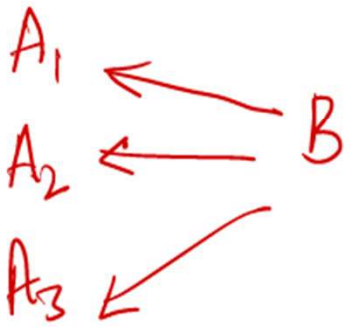
$$A_i \xrightarrow{P(B|A_i)} B$$

(Causal Model)

$$A_i \xleftarrow{P(A_i|B)} B$$

(Inference)





$\{A_1, A_2, \cdots A_i, \cdots, A_n\}$

$$P(A_i|B) = \frac{P(A_i \cap B)}{P(B)}$$ [Definition of Cond. Prob.]

$$= \frac{P(A_i) \cdot P(B|A_i)}{P(B)}$$ [Multiplication Rule]

$$= \frac{P(A_i) \cdot P(B|A_i)}{\left[P(A_1) \cdot P(B|A_1) + P(A_2) \cdot P(B|A_2) + \cdots + P(A_n) \cdot P(B|A_n)\right]}$$

[Total Prob. Theorem]

Bayes' Rule

- Consider a set of mutually-exclusive and collectively-exhaustive potential causes $\{A_1, A_2, A_3\}$ that result in a certain effect B
- We observe B and wish to infer the cause (out of A_1, A_2, A_3) that made B happen.
- This is possible by evaluating the following probability:

$$\mathbf{P}(A_i|B)$$

Cause A_i is present & you are interested in how likely it is to cause effect B

$P(B|A_i)$

Effect B present. How likely is it that A_i caused it

Revisiting the Queen's Gambit

Example 1.13. You enter a chess tournament where your probability of winning a game is 0.3 against half the players (call them type 1), 0.4 against a quarter of the players (call them type 2), and 0.5 against the remaining quarter of the players (call them type 3). You play a game against a randomly chosen opponent.

$$\mathbf{P}(A_1) = 0.5, \qquad \mathbf{P}(A_2) = 0.25. \qquad \mathbf{P}(A_3) = 0.25.$$

Also, B is the event of winning, and

$$\mathbf{P}(B \mid A_1) = 0.3, \qquad \mathbf{P}(B \mid A_2) = 0.4, \qquad \mathbf{P}(B \mid A_3) = 0.5.$$

Suppose that you win. What is the probability $P(A_1|B)$ that you had an opponent of type 1?

$A_1 = \{$ match a type 1 $\}$

$A_2 = \{$ " " type 2 $\}$

$A_3 = \{$ " " type 3 $\}$

$B = \{$ winning $\}$

$$P(A_1|B) = \frac{P(A_1 \cap B)}{P(B)} = \frac{P(A_1) \cdot P(B|A_1)}{P(B)}$$

$$= \frac{P(A_1) \cdot P(B|A_1)}{P(A_1) \cdot P(B|A_1) + P(A_2) \cdot P(B|A_2) + P(A_3) P(B|A_3)}$$

$P(A_1 \cap B)$ $\quad$ $P(A_2 \cap B)$ $\quad$ $P(A_3 \cap B)$

$$= \frac{0.5 \times 0.3}{0.5 \times 0.3 + 0.25 \times 0.4 + 0.25 \times 0.5}$$

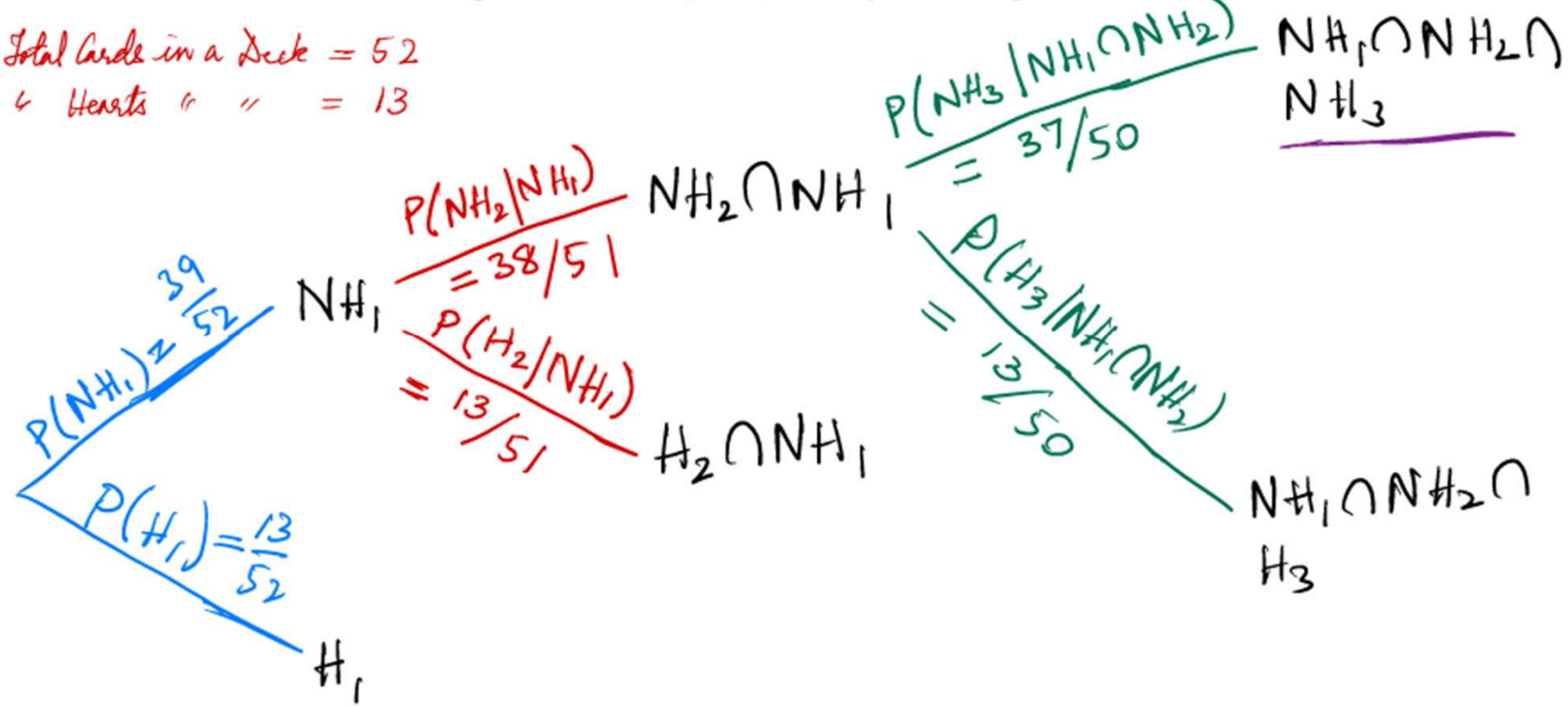
$$= 0.4$$

52 → 51 39 → 38 → Not heart

Example: Put your heart into it

Example 1.10. Three cards are drawn from an ordinary 52-card deck without replacement (drawn cards are not placed back in the deck). We wish to find the probability that none of the three cards is a heart. We assume that at each step, each one of the remaining cards is equally likely to be picked.

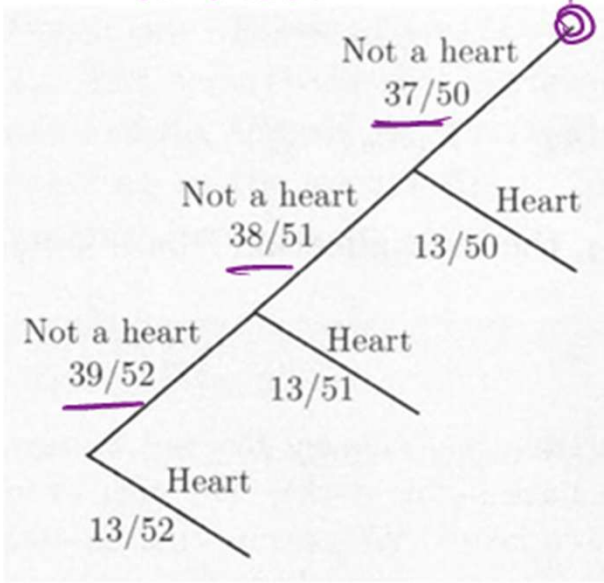
Total Cards in a Deck = 52
4 Hearts " " = 13



Example: Put your heart into it

Example 1.10. Three cards are drawn from an ordinary 52-card deck without replacement (drawn cards are not placed back in the deck). We wish to find the probability that none of the three cards is a heart. We assume that at each step, each one of the remaining cards is equally likely to be picked.

Event of interest



$$P(NH_1 \cap NH_2 \cap NH_3)$$

$$= \frac{39}{52} \times \frac{38}{51} \times \frac{37}{50}$$

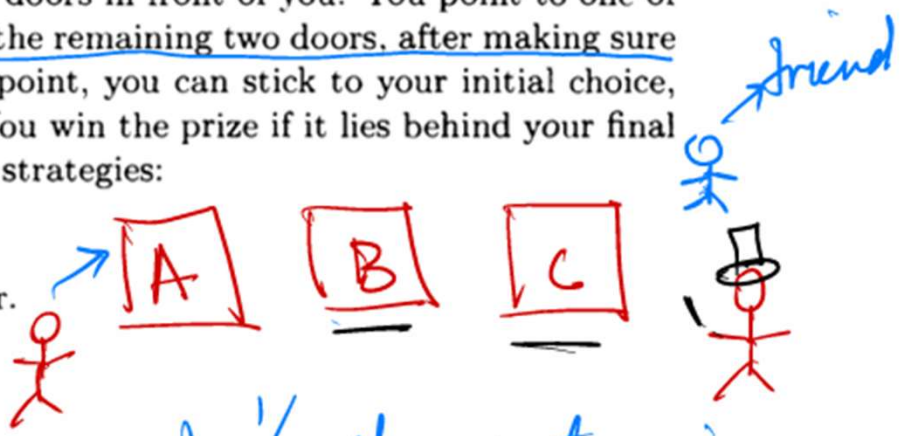
$$= 0.414\ldots$$

Example: The Monty Hall Problem

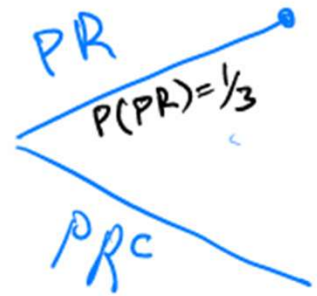
Example 1.12. The Monty Hall Problem. This is a much discussed puzzle, based on an old American game show. You are told that a prize is equally likely to be found behind any one of three closed doors in front of you. You point to one of the doors. A friend opens for you one of the remaining two doors, after making sure that the prize is not behind it. At this point, you can stick to your initial choice, or switch to the other unopened door. You win the prize if it lies behind your final choice of a door. Consider the following strategies:

(a) Stick to your initial choice.

(b) Switch to the other unopened door.



a) 1/3 chance to win
2/3 chance to lose



$P(PR^c) = 2/3$

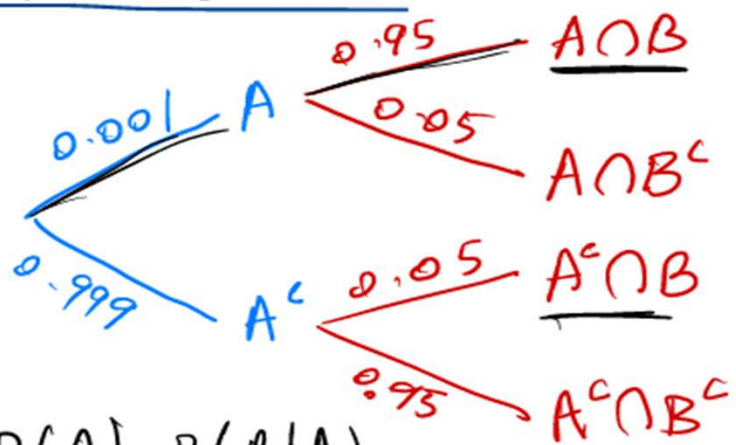
b) 1/3 chance to loss
2/3 chance to win

Example: False-Positives

Example 1.18. The False-Positive Puzzle. A test for a certain rare disease is assumed to be correct 95% of the time: if a person has the disease, the test results are positive with probability 0.95, and if the person does not have the disease, the test results are negative with probability 0.95. A random person drawn from a certain population has probability 0.001 of having the disease. Given that the person just tested positive, what is the probability of having the disease?

$A = \{\text{person gets disease}\}$

$B = \{\text{person testing positive}\}$



$P(B|A) = 0.95$

$P(B^c|A^c) = 0.95$

$P(B^c|A) = 0.05$

$P(B|A^c) = 0.05$

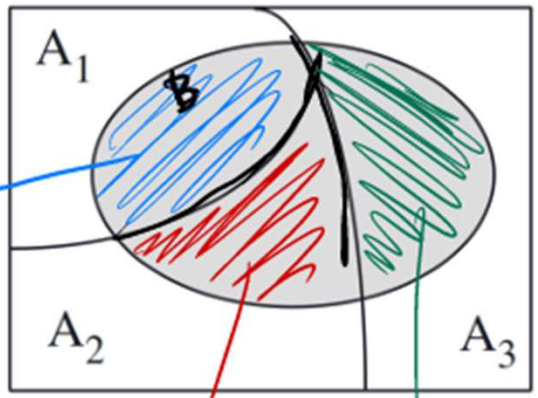
$$P(A|B) = \frac{P(A)\cdot P(B|A)}{P(A)\cdot P(B|A) + P(A^c)\cdot P(B|A^c)}$$

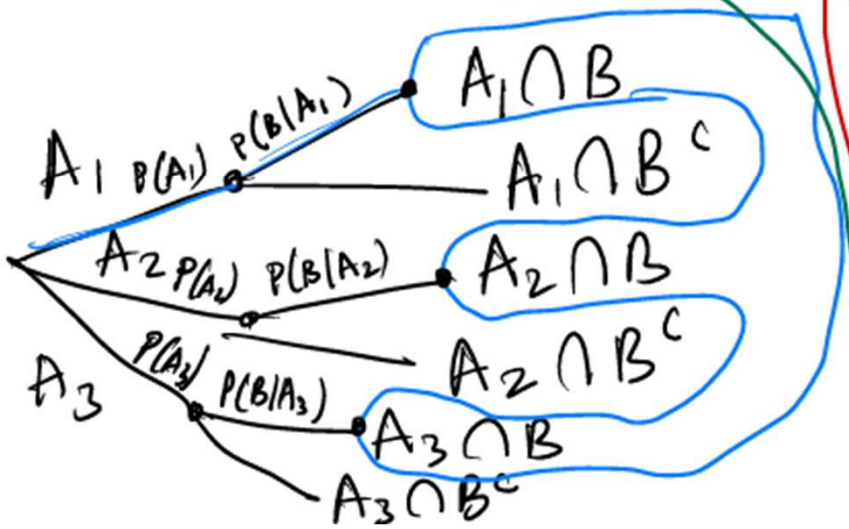
$$P(A|B) = \frac{0.001 \times 0.95}{0.001 \times 0.95 + 0.999 \times 0.05} \approx 0.0187\ldots$$

Total Probability Theorem

- Sample space is partitioned into A_1, A_2, A_3
- How do we calculate $\mathbf{P}(B)$
- $\mathbf{P}(B) = P(A_1 \cap B) + P(A_2 \cap B) +$

$P(A_3 \cap B)$





$P(A_1 \cap B) = P(A_1) \cdot P(B|A_1)$

$P(A_2 \cap B) = P(A_2) \cdot P(B|A_2)$

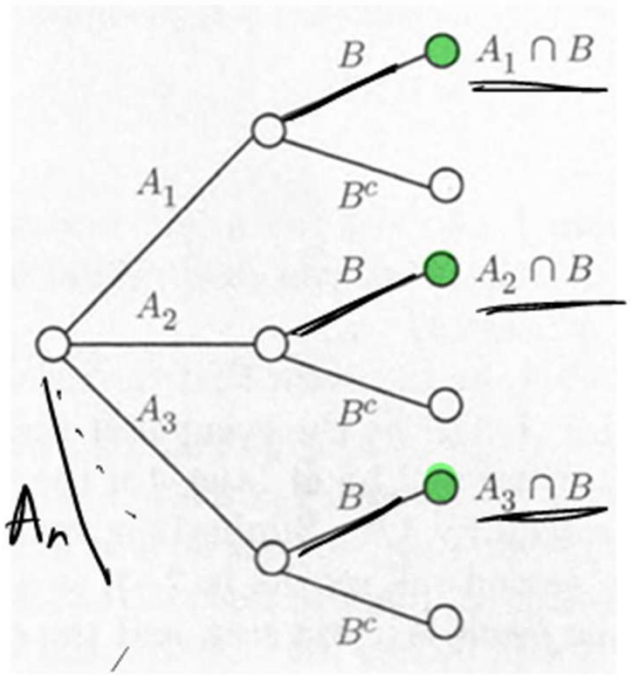
$P(A_3 \cap B) = P(A_3) \cdot P(B|A_3)$

Total Probability Theorem

If $P(A_1) = P(A_2) = P(A_3) = P(A_4) = \cdots = P(A_n)$

$$P(B) = \frac{\sum_i (P(B/A_i))}{n}, \quad i = 1, 2, \cdots, n$$

In more general cases,
$P(B)$ is a weighted avg. of $P(B/A_i)$'s



Independence

① $$\mathbf{P}(A|B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}$$

② $P(A|B) = P(A)$

② → ①

$$P(A) = \frac{P(A \cap B)}{P(B)}$$

$$\boxed{P(A \cap B) = P(A) \cdot P(B)}$$

Necessary Condition for Independence